

Network Traffic Analysis Report using Wireshark

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Environment: Windows 11 Wi-Fi Network

Tool Version: Wireshark

Objective

The objective of this project was to capture and analyze live network traffic using Wireshark to understand communication patterns and introductory network security monitoring techniques.

Tools Used

- Wireshark
- Windows Command Prompt
- Web Browser

Wireshark filters used & Protocols Analyzed

- DNS
- TCP
- TLS/HTTPS
- ICMP
- ARP

Procedure Performed

1. Live Packet Capture

Started live packet capture on the active Wi-Fi interface using Wireshark to monitor real-time network traffic.

No.	Time	Source	Destination	Protocol	Length	Info
423	22.785122	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
424	22.785573	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
425	22.785720	192.168.1.4	23.59.175.10	QUIC	77	Protected Payload (KP0), DCID=0a2092272001c622
426	22.795878	23.59.175.10	192.168.1.4	QUIC	65	Protected Payload (KP0)
427	22.797197	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
428	22.799427	192.168.1.4	23.59.175.10	QUIC	77	Protected Payload (KP0), DCID=0a2092272001c622
429	22.803428	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
430	22.803773	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
431	22.806237	192.168.1.4	23.59.175.10	QUIC	77	Protected Payload (KP0), DCID=0a2092272001c622
432	22.810855	23.59.175.10	192.168.1.4	QUIC	65	Protected Payload (KP0)
433	22.814705	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
434	22.815092	192.168.1.4	23.59.175.10	QUIC	77	Protected Payload (KP0), DCID=0a2092272001c622
435	22.815248	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
436	22.815372	192.168.1.4	23.59.175.10	QUIC	73	Protected Payload (KP0), DCID=0a2092272001c622
437	22.842041	192.168.1.4	23.59.175.10	QUIC	74	Protected Payload (KP0), DCID=0a2092272001c622
438	22.851526	23.59.175.10	192.168.1.4	QUIC	66	Protected Payload (KP0)
439	22.851526	40.104.107.178	192.168.1.4	TCP	54	443 → 61909 [ACK] Seq=5017 Ack=2384 Win=65280 Len=0

> Frame 1: 144 bytes on wire (1152 bits), 144 bytes captured (1152 bits) on interface \Device\NPF_{47508C...}

> Ethernet II, Src: CloudNetwork_06:05:0b (ac:50:de:06:05:0b), Dst: DASANNewtork_85:70:2f (f4:27:56:85:70:2f)

> Internet Protocol Version 4, Src: 192.168.1.4, Dst: 104.18.39.21

> Transmission Control Protocol, Src Port: 59904, Dst Port: 443, Seq: 1, Ack: 1, Len: 90

> Transport Layer Security

0000 f4 27 56 85 70 2f ac 50 de 06 05 0b 08 00 45 00 ..'V.p/.P.....E-
0010 00 82 da 14 40 00 80 06 cf 8d c0 a8 01 04 68 12@.....h-
0020 27 15 ea 00 01 bb 76 0e 95 7d 35 32 6f 80 50 18v...}52o-P-
0030 00 ff 85 7f 00 00 17 03 03 00 55 80 ff 08 4a 72U...Jr
0040 20 66 1e 69 18 20 3f f8 af a8 0d 6f 28 6a a1 92 f-i-?....o(j..
0050 1a 22 db 5e dc 79 92 76 a1 0b 1e dc 05 32 4e d6 .."-y-v.....2N..
0060 76 78 30 f3 e6 73 f3 e1 1f df 19 37 79 c5 e7 ff vx0-s...7y....
0070 70 2d 4d 9f 1b 21 4b 96 60 86 fc 72 bf 12 dd 81 p-M-!K-...p....
0080 de 26 35 51 98 0b a8 02 74 0a 97 d9 81 28 51 87 -85Q.....t....(Q-

2. Traffic Generation

Generated network activity by:

- Visiting websites such as Google, GitHub, and Wikipedia
- Performing ICMP ping requests using Command Prompt

No.	Time	Source	Destination	Protocol	Length	Info
9144	225.220937	192.168.1.4	172.64.155.209	QUIC	1292	Protected Payload (KP0), DCID=01dddf3ebefb303eedf22d3d7efa70bdfcda719
9145	225.220982	192.168.1.4	172.64.155.209	QUIC	788	Protected Payload (KP0), DCID=01dddf3ebefb303eedf22d3d7efa70bdfcda719
9146	225.221157	192.168.1.4	172.64.155.209	QUIC	1286	Protected Payload (KP0), DCID=01dddf3ebefb303eedf22d3d7efa70bdfcda719
9147	225.221222	192.168.1.4	172.64.155.209	QUIC	362	Protected Payload (KP0), DCID=01dddf3ebefb303eedf22d3d7efa70bdfcda719
9148	225.240031	172.64.155.209	192.168.1.4	QUIC	66	Protected Payload (KP0)
9149	225.240216	172.64.155.209	192.168.1.4	QUIC	66	Protected Payload (KP0)
9150	225.240216	172.64.155.209	192.168.1.4	QUIC	66	Protected Payload (KP0)
9151	225.241704	172.64.155.209	192.168.1.4	QUIC	66	Protected Payload (KP0)
9152	225.263669	20.207.73.82	192.168.1.4	TCP	1304	443 → 61793 [ACK] Seq=3320 Ack=2836 Win=72704 Len=1250 [TCP PDU reassembled in 9153]
9153	225.263669	20.207.73.82	192.168.1.4	TLSv1.3	1304	Application Data
9154	225.263669	20.207.73.82	192.168.1.4	TLSv1.3	1304	Application Data
9155	225.263813	192.168.1.4	20.207.73.82	TCP	54	61793 → 443 [ACK] Seq=2836 Ack=7070 Win=65280 Len=0
9156	225.263875	20.207.73.82	192.168.1.4	TLSv1.3	1159	Application Data, Application Data
9157	225.263895	192.168.1.4	20.207.73.82	TCP	54	61793 → 443 [ACK] Seq=2836 Ack=8175 Win=64256 Len=0
9158	225.300448	172.64.155.209	192.168.1.4	QUIC	1242	Protected Payload (KP0)
9159	225.300448	172.64.155.209	192.168.1.4	QUIC	815	Protected Payload (KP0)
9160	225.304070	192.168.1.4	172.64.155.209	QUIC	87	Protected Payload (KP0), DCID=01dddf3ebefb303eedf22d3d7efa70bdfcda719


```

> Frame 1: 144 bytes on wire (1152 bits), 144 bytes captured (1152 bits) on interface \Device\NPF_{47508CC0-0000-0000-0000-000000000000}
> Ethernet II, Src: CloudNetwork_06:05:0b (ac:50:de:06:05:0b), Dst: DASANNetwork_85:70:2f (f4:27:56:85:70:2f)
> Internet Protocol Version 4, Src: 192.168.1.4, Dst: 104.18.39.21
> Transmission Control Protocol, Src Port: 59904, Dst Port: 443, Seq: 1, Ack: 1, Len: 90
> Transport Layer Security
0000  f4 27 56 85 70 2f ac 50 de 06 05 0b 08 00 45 00  :V.p/P .....E-
0010  00 82 da 14 40 00 80 06 cf 8d c0 a8 01 04 68 12  :...@.....h-
0020  27 15 ea 00 01 bb 76 0e 95 7d 35 32 6f 80 50 18  :...v...}52o.P-
0030  00 ff 85 7f 00 00 17 03 03 00 55 00 ff 08 4a 72  :.....[...Jp-
0040  20 66 1e 69 18 20 3f f8 af a8 8d 6f 28 6a a1 92  :f.i. 2. ...o(j--
0050  1a 22 db 5e dc 79 92 76 a1 0b 1e dc 05 32 4e d6  :...y.v .....2N-
0060  76 78 30 f3 e6 73 f3 e1 1f df 19 37 79 c5 e7 ff  :w0+...-7y...
0070  70 2d 44 9f 1b 21 4b 96 60 86 fc 72 bf 12 dd 81  :p:M-1K- .....
0080  de 26 35 51 98 0b a8 02 74 0a 97 d9 81 28 51 87  :85Q....t....(Q-

```

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Microsoft Windows [Version 10.0.26100.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\SNEHALATHA>ping google.com

Pinging google.com [142.251.223.238] with 32 bytes of data:
Reply from 142.251.223.238: bytes=32 time=263ms TTL=117
Reply from 142.251.223.238: bytes=32 time=22ms TTL=117
Reply from 142.251.223.238: bytes=32 time=22ms TTL=117
Reply from 142.251.223.238: bytes=32 time=21ms TTL=117

Ping statistics for 142.251.223.238:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 263ms, Average = 82ms

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3. DNS Analysis

Applied DNS filters to inspect domain resolution requests and responses. Observed how domain names are translated into IP addresses.

No.	Time	Source	Destination	Protocol	Length	Info
17	6.193480	2401:4900:8fcc:8475::	2401:4900:50:9::7b7	DNS	103	Standard query 0x7fa4 A www.msftconnecttest.com
18	6.194571	2401:4900:8fcc:8475::	2401:4900:50:9::7b7	DNS	103	Standard query 0x6afa AAAA www.msftconnecttest.com
19	7.201721	2401:4900:8fcc:8475::	2401:4900:50:9::9	DNS	103	Standard query 0x6afa AAAA www.msftconnecttest.com
20	7.204629	2401:4900:8fcc:8475::	2401:4900:50:9::9	DNS	103	Standard query 0x7fa4 A www.msftconnecttest.com
34	8.205353	192.168.1.4	192.168.1.1	DNS	83	Standard query 0x7fa4 A www.msftconnecttest.com
35	8.207098	192.168.1.4	192.168.1.1	DNS	83	Standard query 0x6afa AAAA www.msftconnecttest.com
36	8.229250	192.168.1.1	192.168.1.4	DNS	193	Standard query response 0x7fa4 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 2...
37	8.232411	192.168.1.1	192.168.1.4	DNS	219	Standard query response 0x6afa AAAA www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 2...
72	9.302290	2401:4900:8fcc:8475::	2401:4900:8fcc:8475::	ICMPv6	151	Destination Unreachable (Address unreachable)
70	9.302890	2401:4900:8fcc:8475::	2401:4900:8fcc:8475::	ICMPv6	151	Destination Unreachable (Address unreachable)
113	10.795832	2401:4900:8fcc:8475::	2401:4900:8fcc:8475::	ICMPv6	151	Destination Unreachable (Address unreachable)
114	10.795832	2401:4900:8fcc:8475::	2401:4900:8fcc:8475::	ICMPv6	151	Destination Unreachable (Address unreachable)
214	20.200173	192.168.1.4	192.168.1.1	DNS	72	Standard query 0x515f HTTPS www.bing.com
215	20.200682	192.168.1.4	192.168.1.1	DNS	72	Standard query 0x8f38 AAAA www.bing.com
216	20.200881	192.168.1.4	192.168.1.1	DNS	72	Standard query 0x1ecd A www.bing.com
217	20.215386	192.168.1.1	192.168.1.4	DNS	254	Standard query response 0x515f HTTPS www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME...
218	20.216425	192.168.1.1	192.168.1.4	DNS	333	Standard query response 0x8f38 AAAA www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME...
219	20.217478	192.168.1.1	192.168.1.4	DNS	337	Standard query response 0x1ecd A www.bing.com CNAME www-www.bing.com.trafficmanager.net CNAME www.bing.com.edgekey.net CNAME e8...


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> Frame 17: 103 bytes on wire (824 bits), 103 bytes captured (824 bits) on interface \Device\NPF_{47508CC0-0000-0000-0000-000000000000}
> Ethernet II, Src: CloudNetwork_06:05:0b (ac:50:de:06:05:0b), Dst: DASANNetwork_85:70:2f (f4:27:56:85:70:2f)
> Internet Protocol Version 6, Src: 2401:4900:8fcc:8475:b5a8:a9fb:8d3c:b7f7, Dst: 2401:4900:50:9::7b7
> User Datagram Protocol, Src Port: 65385, Dst Port: 53
> Domain Name System (query)
0000  f4 27 56 85 70 2f ac 50 de 06 05 0b 06 dd 00 05  :V.p/P .....E-
0010  1e 4e 00 11 11 40 24 01 49 00 8f cc 84 75 b5 a8  :N.1B5 I...u--
0020  a9 fb 8d 3c b7 f7 24 01 49 00 00 50 00 09 00 00  :...$.I-.P....
0030  00 00 00 00 07 b7 ff 69 00 35 00 31 d4 02 7f a4  :.....1-5:1....
0040  01 00 00 01 00 00 00 00 00 03 77 77 7f 0f 6d  :.....www:m
0050  73 66 74 63 6f 6e 6e 65 63 74 74 65 73 74 03 63  :sfctonne cttest c
0060  6f 6d 00 00 01 00 01  :om:....

```

4. TLS/HTTPS Analysis

The figure displays a Wireshark network traffic analysis. The top pane shows a list of captured packets, highlighting a TLSv1.2 session starting at frame 192. The middle pane provides a detailed view of the selected packet (Frame 192), showing it consists of 78 bytes on the wire and was captured on interface \Device\NPF_{47508CCC-...}. The bottom pane shows the expanded details of the TLS record layer, indicating it's a TLS 1.2 connection (0x0303) encrypted with Application Data. The hex dump and ASCII representation of the packet payload are also visible.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.4	104.18.39.21	TLSv1.2	144	Application Data
7	0.854426	104.18.39.21	192.168.1.4	TLSv1.2	133	Application Data
56	8.958906	192.168.1.4	104.18.39.21	TLSv1.2	144	Application Data
75	9.296530	104.18.39.21	192.168.1.4	TLSv1.2	134	Application Data
117	10.960569	192.168.1.4	104.18.39.21	TLSv1.2	144	Application Data
122	11.119387	104.18.39.21	192.168.1.4	TLSv1.2	134	Application Data
129	12.458130	104.18.39.21	192.168.1.4	TLSv1.2	78	Application Data
130	12.458286	192.168.1.4	104.18.39.21	TLSv1.2	82	Application Data
187	17.570845	192.168.1.4	104.18.39.21	TLSv1.2	144	Application Data
192	17.578528	104.18.39.21	192.168.1.4	TLSv1.2	78	Application Data
203	17.579952	192.168.1.4	104.18.39.21	TLSv1.2	82	Application Data
208	18.309605	104.18.39.21	192.168.1.4	TLSv1.2	134	Application Data
228	20.551444	192.168.1.4	23.59.175.10	TLSv1.3	941	Client Hello (SNI=www.bing.com)
230	20.571111	23.59.175.10	192.168.1.4	TLSv1.3	318	Server Hello, Change Cipher Spec, Application Data, Application Data
231	20.571611	192.168.1.4	23.59.175.10	TLSv1.3	134	Change Cipher Spec, Application Data
232	20.571960	192.168.1.4	23.59.175.10	TLSv1.3	146	Application Data
234	20.572523	192.168.1.4	23.59.175.10	TLSv1.3	1011	Application Data
239	20.572770	192.168.1.4	23.59.175.10	TLSv1.3	571	Application Data

```

> Frame 192: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface \Device\NPF_{47508CCC-...}
> Ethernet II, Src: DASEANetw0rk_85:70:31 (f4:27:56:85:70:31), Dst: CloudNetwork_06:05:0b (ac:50:d6:b6:05:0b)
> Internet Protocol Version 4, Src: 104.18.39.21, Dst: 192.168.1.4
> Transmission Control Protocol, Src Port: 443, Dst Port: 56872, Seq: 1, Ack: 1, Len: 24
< Transport Layer Security
  < TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Content Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 19
    Encrypted Application Data: 4b027c31fb8c2b5d098e054adb187a9454ddc
    [Application Data Protocol: Hypertext Transfer Protocol]
  
```

```

0000 ac 50 d6 06 05 0b f4 27 56 85 70 31 08 00 45 88 .P.....V.p1.E-
0010 00 40 22 83 40 00 3a 06 cc d9 68 12 27 15 c0 a8 @""@:-h'
0020 01 04 01 bb de 28 0a 45 f7 50 0d 70 ec b6 50 18 .....(E.P.p-
0030 00 12 c2 e8 00 00 17 03 03 00 13 4b 02 7c 31 fb ..c2.e8.00..170300134b027c31fb
0040 8c 2b 50 d9 8e 05 4a db b1 87 a9 45 dd cd      2b.50.d9.8e.05.4a.db.b1.87.a9.45.dd.cd
  
```

No.	Time	Source	Destination	Protocol Length Info
228	20.551444	192.168.1.4	23.59.175.10	TLSv1.3 941 Client Hello (SNI=www.bing.com)
230	20.571111	23.59.175.10	192.168.1.4	TLSv1.3 318 Server Hello, Change Cipher Spec, Application Data, Application Data
284	20.795409	192.168.1.4	23.59.175.10	TLSv1.3 877 Client Hello (SNI=www.bing.com)
289	20.850789	23.59.175.10	192.168.1.4	TLSv1.3 318 Server Hello, Change Cipher Spec, Application Data, Application Data
306	22.121220	192.168.1.4	23.59.175.10	QUIC 1292 Initial, DCID=49caf1e69d88be54, PKN: 2, CRYPTO, CRYPTO, PADDING, PING, PING, PADDING, CRYPTO, PADDING, CRYPTO, PADDING, CRY-
326	22.221319	23.59.175.10	192.168.1.4	QUIC 1292 Initial, SCID=0a209227208f1c622, PKN: 3, CRYPTO, PADDING
357	22.345880	<0>	40.104.107.178	TLSv1.3 991 Client Hello (SNI=substrate.office.com)
358	22.535713	40.104.107.178	192.168.1.4	TLSv1.3 1304 Server Hello, Change Cipher Spec
643	49.108502	192.168.1.4	20.42.65.88	TLSv1.3 611 Client Hello (SNI=self.events.data.microsoft.com)
645	52.180452	20.42.65.88	192.168.1.4	TLSv1.3 1304 Server Hello, Change Cipher Spec
695	57.985746	192.168.1.4	23.59.175.10	QUIC 1292 Initial, DCID=9f12674da59f19ef, PKN: 2, PADDING, CRYPTO, PING, PING, CRYPTO, PING, PADDING, PING, PADDING, CRYPTO, PADDING,
716	58.010984	23.59.175.10	192.168.1.4	QUIC 1292 Initial, SCID=02fa9366590c622, PKN: 3, CRYPTO, PADDING
935	72.336571	192.168.1.4	23.53.240.136	TLSv1.3 535 Client Hello (SNI=sadownload.mcafee.com)
938	72.344542	23.53.240.136	192.168.1.4	TLSv1.3 1304 Server Hello, Change Cipher Spec, Application Data
951	72.605152	192.168.1.4	23.53.240.136	TLSv1.3 599 Client Hello (SNI=sadownload.mcafee.com)
956	72.608080	23.53.240.136	192.168.1.4	TLSv1.3 1304 Server Hello, Change Cipher Spec, Application Data
1001	72.643695	192.168.1.4	142.251.222.131	QUIC 1292 Initial, DCID=07cb65f420eb3bc0, PKN: 1, PING, PING, CRYPTO, PING, PING, CRYPTO, PING, PING, CRYPTO, CRYPTO
1002	72.643821	192.168.1.4	142.251.222.131	QUIC 1292 Initial, DCID=07cb65f420eb3bc0, PKN: 2, PING-CRYPTO-PING.PADDING-PING.CRYPTO
> Frame 358: 1304 bytes on wire (10432 bits), 1304 bytes captured (10432 bits) on interface \Device\NPF{...}				
> Ethernet II, Src: DASENetworkK_85:70:31 (F4:27:56:85:70:31), Dst: CloudNetwrok_06:05:0b (ac:c5:d0:e6:05:				
> Internet Protocol Version 4, Src: 40.104.107.178, Dst: 192.168.1.4				
> Transmission Control Protocol, Src Port: 443, Dst Port: 61909, Seq: 1, Ack: 938, Len: 1250				
Transport Layer Security				
TLSv1.3 Record Layer: Handshake Protocol: Server Hello				
Content Type: Handshake (22)				
Version: TLS 1.2 (0x0303)				
Length: 187				
Handshake Protocol: Server Hello				
TLSv1.3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec				
Content Type: Change Cipher Spec (20)				
Version: TLS 1.2 (0x0303)				
Length: 1				
Change Cipher Spec Message				
TLS segment data (1052 bytes)				

5. TCP Handshake Analysis

Identified the TCP three-way handshake process involving SYN, SYN-ACK, and ACK packets used to establish reliable client-server communication sessions.

File Edit Go Capture Analyze Statistics Telephony Tools Help

tcp

No.	Time	Source	Destination	Protocol	Length	Info
322	22.147262	23.59.175.10	192.168.1.4	TCP	60	443 → 61502 [ACK] Seq=15611 Ack=28891 Win=61952 Len=0
344	22.274567	23.59.175.10	192.168.1.4	TLSv1.3	479	Application Data
345	22.274567	23.59.175.10	192.168.1.4	TLSv1.3	85	Application Data
346	22.274707	192.168.1.4	23.59.175.10	TCP	54	61502 → 443 [ACK] Seq=28891 Ack=16067 Win=65024 Len=0
351	22.462266	192.168.1.4	40.104.107.178	TCP	66	61909 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
352	22.477374	40.104.107.178	192.168.1.4	TCP	66	443 → 61909 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1250 WS=256 SACK_PERM
353	22.477526	192.168.1.4	40.104.107.178	TCP	54	61909 → 443 [ACK] Seq=1 Ack=1 Win=262144 Len=0
357	22.518680	192.168.1.4	40.104.107.178	TLSv1.3	991	Client Hello (SNI=substrate.office.com)
358	22.535713	40.104.107.178	192.168.1.4	TLSv1.3	1304	Server Hello, Change Cipher Spec
359	22.535713	40.104.107.178	192.168.1.4	TCP	1304	443 → 61909 [ACK] Seq=1251 Ack=938 Win=64512 Len=1250 [TCP PDU reassembled in 361]
> Frame 352: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{47508CCC-0000-ac-50-de-06-05-0b} f4:27:56:85:70:2f, Dst: CloudNetwork_06:05:0b (ac:50:de:06:05:0b)						
> Ethernet II, Src: DASHNetwork_85:70:2f (f4:27:56:85:70:2f), Dst: CloudNetwork_06:05:0b (ac:50:de:06:05:0b)						
> Internet Protocol Version 4, Src: 40.104.107.178, Dst: 192.168.1.4						
Transmission Control Protocol, Src Port: 443, Dst Port: 61909, Seq: 0, Ack: 1, Len: 0						
Source Port: 443						
Destination Port: 61909						
[Stream index: 12]						
[Stream Packet Number: 2]						
> [Conversation completeness: Complete, WITH_DATA (31)]						
[TCP Segment Len: 0]						
Sequence Number: 0 (relative sequence number)						
Sequence Number (raw): 1779459308						
[Next Sequence Number: 1 (relative sequence number)]						
Acknowledgment Number: 1 (relative ack number)						
Acknowledgment number (raw): 2924716691						
1000 = Header Length: 32 bytes (8)						
> Flags: 0x012 (SYN, ACK)						
Window: 65535						
[Calculated window size: 65535]						
Checksum: 0x0697 [unverified]						
[Checksum Status: Unverified]						
Urgent Pointer: 0						
> Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-C						
> [Timestamps]						
> [SEQ/ACK analysis]						

0000 ac 50 de 06 05 0b f4 27 56 85 70 2f 08 00 45 b8 P.....Vp...E...
0010 00 04 e2 c3 40 00 75 06 0c 82 28 68 6b b2 c9 a8 4...@...{hk...
0020 01 04 01 bf d5 6a 10 64 ec ae 53 a2 93 80 12j...d-S...
0030 ff ff 06 97 00 00 02 04 04 e2 01 03 03 08 01 01
0040 04 02 ..

6. ICMP Analysis

Observed ICMP traffic generated from ping requests and identified network diagnostic communication patterns.

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icmp

No.	Time	Source	Destination	Protocol	Length	Info
7456	158.825198	192.168.1.4	142.251.223.238	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 7457)
7419	157.809544	192.168.1.4	142.251.223.238	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 7422)
7321	156.792546	192.168.1.4	142.251.223.238	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 7324)
7316	155.786975	192.168.1.4	142.251.223.238	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 7318)
7457	158.846975	142.251.223.238	192.168.1.4	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=117 (request in 7456)
7422	157.831790	142.251.223.238	192.168.1.4	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=117 (request in 7419)
7324	156.814797	142.251.223.238	192.168.1.4	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=117 (request in 7321)
7318	156.050318	142.251.223.238	192.168.1.4	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=117 (request in 7316)
9107	225.015937	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8860	223.358628	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8662	216.703283	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8598	215.036583	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8514	213.384099	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8235	206.749661	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8226	205.092593	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
8205	203.433396	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
7544	161.507489	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
7514	159.848740	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
7424	158.101022	192.168.1.1	192.168.1.4	ICMP	138	Destination unreachable (Port unreachable)
> Frame 7422: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\NPF_{47508CCC-0000-ac-50-de-06-05-0b} f4:27:56:85:70:2f, Dst: CloudNetwork_06:05:0b (ac:50:de:06:05:0b)						
> Ethernet II, Src: DASHNetwork_85:70:2f (f4:27:56:85:70:2f), Dst: CloudNetwork_06:05:0b (ac:50:de:06:05:0b)						
> Internet Protocol Version 4, Src: 142.251.223.238, Dst: 192.168.1.4						
Internet Control Message Protocol						
Type: 0 (Echo (ping) reply)						
Code: 0						
Checksum: 0x5558 [correct]						
[Checksum Status: Good]						
Identifier (BE): 1 (0x0001)						
Identifier (LE): 256 (0x0100)						
Sequence Number (BE): 3 (0x0003)						
Sequence Number (LE): 768 (0x0300)						
[Request frame: 7419]						
[Response time: 22.246 ms]						
> Data (32 bytes)						

0000 ac 50 de 06 05 0b f4 27 56 85 70 2f 08 00 45 00 P.....V.p...E..

0010 00 3c 00 00 00 00 75 01 15 2b be fb df ea c0 a8 <...u...+...

0020 01 04 00 00 55 58 09 01 09 03 b1 62 63 64 65 66UX...abdefg

0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76 ghijklmn opqrstuv

0040 77 61 62 63 64 65 66 67 68 69 wabdefgh i

7. ARP Analysis

Observed how ARP traffic can potentially be abused in ARP spoofing attacks within local networks.

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arp						
No.	Time	Source	Destination	Protocol	Length	Info
16	5.801544	DASANNewtork_85:70:...	Broadcast	ARP	42	Who has 192.168.1.2? Tell 192.168.1.1

> Frame 16: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{47508CCC-2}		0000	ff ff ff ff ff f4 27 56 85 70 2f 00 06 00 01	' V.p/...
> Ethernet II, Src: DASANNewtork_85:70:2f (f4:27:56:85:70:2f), Dst: Broadcast (ff:ff:ff:ff:ff:ff)		0010	00 00 06 04 00 01 f4 27 56 85 70 2f c0 a8 01 01	' V.p/...
v Address Resolution Protocol (request)		0020	00 00 00 00 00 00 c0 a8 01 02	
Hardware type: Ethernet (1)				
Protocol type: IPv4 (0x0800)				
Hardware size: 6				
Protocol size: 4				
Opcode: request (1)				
Sender MAC address: DASANNewtork_85:70:2f (f4:27:56:85:70:2f)				
Sender IP address: 192.168.1.1				
Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)				
Target IP address: 192.168.1.2				

Key Observations

- DNS requests were generated immediately after visiting websites.
- TLS packets contained encrypted application data.
- ICMP packets showed request-reply communication patterns.
- ARP traffic was limited to local subnet communication.

Findings

- Successfully captured live network traffic.
- Observed DNS query and response behaviour.
- Analyzed encrypted TLS communication.
- Identified TCP handshake packets.
- Observed ICMP communication messages.
- Reviewed ARP traffic and understood ARP spoofing concepts.

Conclusion

This project provided hands-on experience in packet capture and protocol analysis using Wireshark. The investigation improved practical understanding of TCP/IP communication, encrypted traffic behaviour, DNS resolution, and network monitoring techniques commonly used in SOC and incident response environments.

Skills Gained

- Packet Analysis
- Wireshark Filtering
- Protocol Inspection
- TCP/IP Fundamentals
- DNS Analysis
- Traffic Monitoring
- Basic Threat Detection
- Incident Documentation